SOLVED EXAMPLES

Ex. 1 : Write $(1+2i)\,(1+3i)\,(2+i)^{-1}$ in the form $a+ib$

Solution :

$$(1+2i)\,(1+3i)\,(2+i)^{-1} = \frac{(1+2i)(1+3i)}{2+i}$$

$$= \frac{1+3i+2i+6i^2}{2+i} = \frac{-5+5i}{2+i} \times \frac{2-i}{2-i}$$

$$= \frac{-10+5i+10i-5i^2}{4-i^2} = \frac{-5+15i}{4+1} \quad (\because i^2 = -1)$$

$$= \frac{-5+15i}{5} = -1 + 3i$$

Ex. 2 : (Activity) Express $\frac{1}{i} + \frac{2}{i^2} + \frac{3}{i^3} + \frac{5}{i^4}$ in the form of $(a + ib)$.

Solution : $i^2 = -1$, $i^3 = -i$, $i^4 = 1$

$\frac{1}{i} = \square$, $\frac{1}{i^2} = \square$, $\frac{1}{i^3} = \square$, $\frac{1}{i^4} = \square$

$\therefore \quad \frac{1}{i} + \frac{2}{i^2} + \frac{3}{i^3} + \frac{5}{i^4}$

$= 1\,(\square) + 2\,(\square) + 3\,(\square) + 5\,(\square)$

$= \square + i\,\square$

$\therefore \quad a = \square, \; b = \square$

Ex. 3 : If a and b are real and $(i^4+3i)a + (i-1)b + 5i^3 = 0$, find a and b.

Solution : $(i^4+3i)a + (i-1)b + 5i^3 = 0+0i$

i.e. $(1+3i)a + (i-1)b - 5i = 0+0i$

$\therefore\ a + 3ai + bi - b - 5i = 0+0i$

i.e. $(a-b) + (3a+b-5)i = 0+0i$

By equality of complex numbers, we get

$a-b = 0$ and $3a+b-5 = 0$

$\therefore\ a=b$ and $3a+b = 5$

$\therefore\ 3a+a = 5$

$\therefore\ \ 4a = 5$

$\therefore\ a = \frac{5}{4}$

$\therefore\ a = b = \frac{5}{4}$

Ex. 4 : If $x + 2i + 15i^6y = 7x + i^3\,(y+4)$ find $x + y$, given that $x, y \in R$.

Solution :

$x + 2i + 15i^6y = 7x + i^3\,(y+4)$

$\therefore\ x + 2i - 15y = 7x - (y+4)\,i$

$(\because i^6 = -1,\ i^3 = -i)$

$\therefore\ x - 15y + 2i = 7x - (y+4)\,i$

Equating real and imaginary parts, we get

$x - 15y = 7x$ and $2 = -(y+4)$

$\therefore\ -6x - 15y = 0$ .... (i) $\quad y+6 = 0$ ....(ii)

$\therefore\ y = -6,\ x = 15$ [Solving (i) and (ii)]

$\therefore\ x + y = 15 - 6 = 9$

Ex. 5 : Show that $\left(\frac{\sqrt{3}}{2} + \frac{i}{2}\right)^3 = i$.

Solution :

$$\text{L.H.S.} = \left(\frac{\sqrt{3}}{2} + \frac{i}{2}\right)^3 = \left(\frac{\sqrt{3}+i}{2}\right)^3$$

$$= \frac{\left(\sqrt{3}\right)^3 + 3\left(\sqrt{3}\right)^2 i + 3\sqrt{3}i^2 + (i)^3}{(2)^3}$$

$$= \frac{3\sqrt{3} + 9i - 3\sqrt{3} - i}{8}$$

$$= \frac{8i}{8}$$

$$= i$$

$$= \text{R.H.S.}$$

EXERCISE 1.1

1) Simplify :

i) $\sqrt{-16} + 3\sqrt{-25} + \sqrt{-36} - \sqrt{-625}$

ii) $4\sqrt{-4} + 5\sqrt{-9} - 3\sqrt{-16}$

2) Write the conjugates of the following complex numbers

i) $3+i$ ii) $3-$ iii) $-\sqrt{5}-\sqrt{7}i$

iv) $-\sqrt{-5}$ v) $5i$ vi) $\sqrt{5}-i$

vii) $\sqrt{2}+\sqrt{3}\,i$ viii) $\cos\theta + i\sin\theta$

3) Find a and b if

i) $a + 2b + 2ai = 4 + 6i$

ii) $(a-b) + (a+b)i = a + 5i$

iii) $(a+b)(2 + i) = b + 1 + (10 + 2a)i$

iv) $abi = 3a - b + 12i$

v) $\dfrac{1}{a+ib} = 3 - 2i$

vi) $(a + ib)(1 + i) = 2 + i$

4) Express the following in the form of $a+ib$, $a, b \in R$ $i=\sqrt{-1}$. State the values of a and b.

i) $(1+2i)(-2+i)$ ii) $(1+i)(1-i)^{-1}$

iii) $\dfrac{i(4+3i)}{(1-i)}$ iv) $\dfrac{(2+i)}{(3-i)(1+2i)}$

v) $\left(\dfrac{1+i}{1-i}\right)^2$ vi) $\dfrac{3+2i}{2-5i}+\dfrac{3-2i}{2+5i}$

vii) $(1+i)^{-3}$ viii) $\dfrac{2+\sqrt{-3}}{4+\sqrt{-3}}$

ix) $(-\sqrt{5}+2\sqrt{-4})+(1-\sqrt{-9})+(2+3i)(2-3i)$

x) $(2+3i)(2-3i)$ xi) $\dfrac{4i^8-3i^9+3}{3i^{11}-4i^{10}-2}$

5) Show that $(-1+\sqrt{3}i)^3$ is a real number.

6) Find the value of $\left(3+\dfrac{2}{i}\right)(i^6-i^7)(1+i^{11})$

7) Evaluate the following :

i) i^{35} ii) i^{888} iii) i^{93} iv) i^{116}

v) i^{403} vi) $\dfrac{1}{i^{58}}$ vii) i^{-888}

viii) $i^{30}+i^{40}+i^{50}+i^{60}$

8) Show that $1 + i^{10} + i^{20} + i^{30}$ is a real number.

9) Find the value of

i) $i^{49}+i^{68}+i^{89}+i^{110}$

ii) $i+i^2+i^3+i^4$

10) Simplify : $\dfrac{i^{592}+i^{590}+i^{588}+i^{586}+i^{584}}{i^{582}+i^{580}+i^{578}+i^{576}+i^{574}}$

11) Find the value of $1 + i^2 + i^4 + i^6 + i^8 + \ldots + i^{20}$

12) Show that $1 + i^{10} + i^{100} - i^{1000} = 0$.

13) Is $(1 + i^{14} + i^{18} + i^{22})$ a real number? Justify your answer.

14) Evaluate : $\left(i^{37}+\dfrac{1}{i^{67}}\right)$

15) Prove that $(1+i)^4 \times \left(1+\dfrac{1}{i}\right)^4 = 16$.

16) Find the value of $\dfrac{i^6+i^7+i^8+i^9}{i^2+i^3}$

17) If $a = \dfrac{-1+\sqrt{3}i}{2}$, $b = \dfrac{-1-\sqrt{3}i}{2}$ then show that $a^2 = b$ and $b^2 = a$.

18) If $x+iy = (a+ib)^3$, show that $\dfrac{x}{a}+\dfrac{y}{b} = 4(a^2-b^2)$

19) If $\dfrac{a+3i}{2+ib} = 1-i$, show that $(5a-7b) = 0$.

20) If $x+iy = \sqrt{\dfrac{a+ib}{c+id}}$, prove that $(x^2+y^2)^2 = \dfrac{a^2+b^2}{c^2+d^2}$

21) If $(a+ib) = \dfrac{1+i}{1-i}$, then prove that $(a^2+b^2) = 1$.

22) Show that $\left(\dfrac{\sqrt{7}+i\sqrt{3}}{\sqrt{7}-i\sqrt{3}}+\dfrac{\sqrt{7}-i\sqrt{3}}{\sqrt{7}+i\sqrt{3}}\right)$ is real.

23) If $(x+iy)^3 = y+vi$ the show that

$\left(\dfrac{y}{x}+\dfrac{v}{y}\right) = 4(x^2-y^2)$

24) Find the value of x and y which satisfy the following equations $(x, y \in R)$

i) $(x+2y) + (2x-3y)\,i + 4i = 5$

ii) $\frac{x+1}{1+i} + \frac{y-1}{1-i} = i$

iii) $\frac{(x+iy)}{2+3i} + \frac{2+i}{2-3i} = \frac{9}{13}(1+i)$

iv) If $x(1+3i) + y(2-i) - 5 + i^3 = 0$, find $x+y$

v) If $x+2i+15i^6y = 7x+i^3(y+4)$, find $x+y$

1.3 Square root of a complex number :

Consider $z = x+iy$ be any complex number

Let $\sqrt{x+iy} = a+ib$, $a, b \in R$

On squaring both the sides, we get

$x+iy = (a+ib)^2$

$x+iy = (a^2-b^2) + (2ab)\,i$

Equating real and imaginary parts, we get

$x = (a^2-b^2)$ and $y = 2ab$

Solving these equations simultaneously, we can get the values of a and b.

Solved Examples:

Ex.1 : Find the square root of $6+8i$.

Solution :

Let, $\sqrt{6+8i} = a+ib$ $(a, b \in R)$

On squaring both the sides, we get

$6+8i = (a+ib)^2$

$\therefore 6+8i = (a^2-b^2) + (2ab)\,i$

Equating real and imaginary parts, we have

$6 = a^2-b^2$ .... (1)

$8 = 2ab$ .... (2)

$\therefore a = \frac{4}{b}$

$\therefore$ (1) becomes $6 = \left(\frac{4}{b}\right)^2 - b^2$

i.e. $6 = \frac{16}{b^2} - b^2$

$\therefore b^4+6b^2-16 = 0$ i.e. $(b^2)^2 + 6b^2 - 16 = 0$

put $b^2 = m$

$\therefore m^2+6m-16 = 0$

$\therefore (m+8)(m-2) = 0$

$\therefore m = -8$ or $m = 2$

i.e. $b^2 = -8$ or $b^2 = 2$

but b is a real number $\therefore b^2 \neq -8$

So, $b^2 = 2$ $\therefore b = \pm\sqrt{2}$

For, $b = \sqrt{2}$, $a = 2\sqrt{2}$

$\therefore \sqrt{6+8i} = 2\sqrt{2} + \sqrt{2}\,i = \sqrt{2}(2+i)$

For, $b = -\sqrt{2}$, $a = -2\sqrt{2}$

$\therefore \sqrt{6+8i} = -2\sqrt{2} - \sqrt{2}\,i = -\sqrt{2}(2+i)$

$\therefore \sqrt{6+8i} = \pm\sqrt{2}(2+i)$

Ex. 2 : Find the square root of $3 - 4i$

Solution :

Let $\sqrt{3-4i} = a+ib$ $a, b \in R$

On squaring both the sides, we have

$3 - 4i = (a+ib)^2$

$\therefore 3 - 4i = (a^2-b^2) + (2ab)\,i$

Equating real and imaginary parts, we have

$a^2-b^2 = 3$, $2ab = -4$

As $(a^2+b^2)^2 = (a^2-b^2)^2 + (2ab)^2$

$(a^2+b^2)^2 = 3^2 + (-4)^2 = 9 + 16 = 25$

$(a^2+b^2)^2 = 5^2$

$\therefore a^2+b^2 = 5$

Solving $a^2+b^2 = 5$ and $a^2-b^2 = 3$ we get

$2a^2 = 8$

$a^2 = 4$

$\therefore a = \pm 2$

For $a = 2$;

$b = \frac{-4}{2a} = \frac{-4}{2(2)} = -1$

For, $a = -2$, $b = \frac{-4}{2(-2)} = 1$

$\therefore \sqrt{3-4i} = 2 - i$ or $-2 + i$

1.4 Fundamental Theorem of Algebra :

'A polynomial equation with real coefficients has at least one root' in C.

or 'A polynomial equation with complex coefficients and of degree n has n complex roots'.

1.4.1 Solution of a Quadratic Equation in complex number system :

Let the given equation be $ax^2 + bx + c = 0$ where $a, b, c \in R$ and $a \neq 0$

$\therefore$ The solution of this quadratic equation is given by

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Hence, the roots of the equation $ax^2 + bx + c = 0$ are $\frac{-b + \sqrt{b^2 - 4ac}}{2a}$ and $\frac{-b - \sqrt{b^2 - 4ac}}{2a}$

The expression $(b^2 - 4ac) = D$ is called the discriminant.

If $D < 0$ then the roots of the given quadratic equation are complex.

Note : If $p + iq$ is the root of equation $ax^2 + bx + c = 0$ where $a, b, c \in R$ and $a \neq 0$ then $p - iq$ is also a solution of the given equation. Thus, complex roots occur in conjugate pairs.

Solved Examples :

Ex. 1 : Solve $x^2 + x + 1 = 0$

Solution : Given equation is $x^2 + x + 1 = 0$

Comparing with $ax^2 + bx + c = 0$ we get

$a = 1, \quad b = 1, c = 1$

These roots are given by

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-1 \pm \sqrt{-3}}{2}$$

$$= \frac{-1 \pm \sqrt{3}\, i}{2}$$

$\therefore$ Roots are $\frac{-1 + \sqrt{3}\, i}{2}$ and $\frac{-1 - \sqrt{3}\, i}{2}$

Ex. 2 : Solve $x^2 - (2\sqrt{3} + 3i)\, x + 6\sqrt{3}\, i = 0$

Solution : Given equation is

$x^2 - (2\sqrt{3} + 3i)\, x + 6\sqrt{3}\, i = 0$

The method of finding the roots of

$ax^2 + bx + c = 0$, is applicable even if a, b, c are complex numbers. where $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

Here, $a = 1, b = -(2\sqrt{3} + 3i), c = 6\sqrt{3}\, i$

$b^2 - 4ac = [-(2\sqrt{3} + 3i)]^2 - 4 \times 1 \times 6\sqrt{3}\, i$

$= 12 - 9 + 12\sqrt{3}\, i - 24\sqrt{3}\, i$

$= 3 - 12\sqrt{3}\, i$

$= 3\,(1 - 4\sqrt{3}\, i)$

So, the given equation has complex roots. These roots are given by

$$x = \frac{\left(2\sqrt{3} + 3i\right) \pm \sqrt{3(1 - 4\sqrt{3}\, i)}}{2}$$

Now, we shall find $\sqrt{1 - 4\sqrt{3}\, i}$

Let $a + ib = \sqrt{1 - 4\sqrt{3}\, i}$

$\therefore \quad a^2 - b^2 + 2iab = 1 - 4\sqrt{3}\, i$

$\therefore \quad a^2 - b^2 = 1$ and $2ab = -4\sqrt{3}$

$a^2 - b^2 = 1$ and $ab = -2\sqrt{3}$

Consider $(a^2 + b^2)^2 = (a^2 - b^2)^2 + 4\, a^2 b^2$

$= 1 + 4\,(12) = 49$

$\therefore \quad a^2 + b^2 = 7 \quad ...(1)$

and also $a^2 - b^2 = 1 \quad ...(2)$

Solving (1) and (2)

$2a^2 = 8 \qquad \therefore \quad a^2 = 4$

$\therefore \quad a = \pm 2$ and $b = \pm\sqrt{3}$

$\therefore$ We have four choices

$a = 2, \qquad b = \sqrt{3} \qquad ... (3)$

$a = 2, \qquad b = -\sqrt{3} \qquad ... (4)$

$a = -2, \quad b = \sqrt{3}$ ... (5)

$a = -2, \quad b = -\sqrt{3}$ ... (6)

With this combination, the condition $ab = -2\sqrt{3}$ should also be satisfied.

We can verify the values of a and b given by (4) and (5) satisfy the condition and that from (3) and (6) do not satisfy the condition.

Hence, $a = 2, \quad b = -\sqrt{3}$

and $a = -2, \quad b = \sqrt{3}$

$\therefore \sqrt{1-4\sqrt{3}\,i} = \pm(2-\sqrt{3}\,i)$

$\therefore$ The roots are given by

$$x = \frac{\left(2\sqrt{3}+3i\right) \pm \sqrt{3}\left(2-\sqrt{3}\,i\right)}{2}$$

So, the roots are $2\sqrt{3}$ and $3i$.

Ex. 3 : Find the value of $x^3 - x^2 + 2x + 10$ when $x = 1 + \sqrt{3}\,i$.

Solution : Since $x = 1 + \sqrt{3}\,i$

$\therefore x - 1 = \sqrt{3}\,i$

squaring both sides, we get

$(x-1)^2 = (\sqrt{3}\,i)^2$

$\therefore x^2 - 2x + 1 = 3i^2$

i.e. $x^2 - 2x + 1 = -3$

$\therefore x^2 - 2x = -4$ ..... (I)

$x^3 - x^2 + 2x + 10 = x^3 - (x^2 - 2x) + 10$ (By I)

$= x^3 - (-4) + 10 = x^3 + 14$

$= (1 + \sqrt{3}\,i)^3 + 14$

$= 1 + 3\sqrt{3}\,i - 9 - 3\sqrt{3}\,i + 14$

$= 6$

Ex. 4 : If $x = -5 + 2\sqrt{-4}$, find the value of $x^4 + 9x^3 + 35x^2 - x + 64$.

Solution : $x = -5 + 2\sqrt{-4} = -5 + 2(2i) = -5 + 4i$

Let $P(x) = x^4 + 9x^3 + 35x^2 - x + 64$

Form the quadratic expression $q(x)$ with real coefficients and roots $\alpha = -5+4i$ and $\overline{\alpha} = -5-4i$.

$\alpha\,\overline{\alpha} = (-5+4i)(-5-4i) = 25 + 16 = 41$

$\alpha + \overline{\alpha} = (-5+4i) + (-5-4i) = -10$

Therefore $q(x) = x^2 + 10x + 41$

Now $q(\alpha) = \alpha^2 + 10\alpha + 41 = 0$ ...(1)

We divide the given polynomial $p(x)$ by $q(x)$.

$$\begin{array}{r|l} & x^2 - x + 4 \\ \hline x^2+10x+41 & x^4 + 9x^3 + 35x^2 - x + 64 \\ & x^4 + 10x^3 + 41x^2 \\ \hline & -x^3 + 6x^2 - x \\ & -x^3 - 10x^2 - 41x \\ \hline & 4x^2 + 40x + 64 \\ & 4x^2 + 40x + 164 \\ \hline & -100 \end{array}$$

$\therefore \quad p(x) = (x^2 + 10x + 41)(x^2 - x + 4) - 100$

$\therefore \quad p(\alpha) = q(\alpha)(\alpha^2 - \alpha + 4) - 100$

$\therefore \quad p(\alpha) = 0 - 100 = -100$ (By I)

EXERCISE 1.2

1) Find the square root of the following complex numbers

i) $-8-6i$ ii) $7+24i$ iii) $1+4\sqrt{3}\,i$

iv) $3+2\sqrt{10}\,i$ v) $2(1-\sqrt{3}\,i)$

2) Solve the following quadratic equations.

i) $8x^2 + 2x + 1 = 0$

ii) $2x^2 - \sqrt{3}\,x + 1 = 0$

iii) $3x^2 - 7x + 5 = 0$

iv) $x^2 - 4x + 13 = 0$

3) Solve the following quadratic equations.

i) $x^2 + 3ix + 10 = 0$

ii) $2x^2 + 3ix + 2 = 0$

iii) $x^2 + 4ix - 4 = 0$

iv) $ix^2 - 4x - 4i = 0$

4) Solve the following quadratic equations.

i) $x^2 - (2+i)\,x - (1-7i) = 0$

ii) $x^2 - (3\sqrt{2} + 2i)\,x + 6\sqrt{2}\,i = 0$

iii) $x^2 - (5-i)\,x + (18+i) = 0$

iv) $(2 + i)x^2 - (5-i)\,x + 2\,(1-i) = 0$

5) Find the value of

iii) $x^3 - x^2 + x + 46$, if $x = 2+3i$.

iv) $2x^3 - 11x^2 + 44x + 27$, if $x = \dfrac{25}{3-4i}$.

v) $x^3 + x^2 - x + 22$, if $x = \dfrac{5}{1-2i}$.

vi) $x^4 + 9x^3 + 35x^2 - x + 4$, if $x = -5 + \sqrt{-4}$.

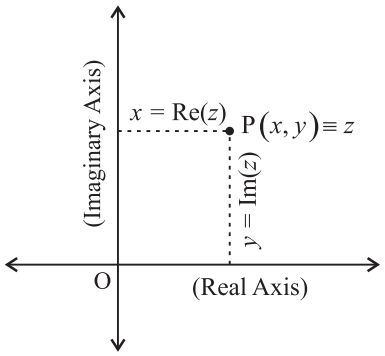
vii) $2x^4 + 5x^3 + 7x^2 - x + 41$, if $x = -2 - \sqrt{3}\,i$.

1.5 Argand Diagram or Complex Plane :

A complex number $z = x + iy$, $x, y \in R$ and $i = \sqrt{-1}$ is expressed as a point in the plane whose co-ordinates are ordered pair (x, y). Jean Robert Argand used the one to one correspondence between a complex number and the points in a the plane.

Let $z = x + iy$ be a complex number.

Then the point $P(x, y)$ represents the complex number $z = x + iy$ (fig.1.2) i.e. $x + iy \equiv (x, y)$, $x = Re(z)$ is represented on the X-axis. So, X-axis is called the real axis. Similarly, $y = Im(z)$ is represented on the Y-axis, so the Y-axis is called the **imaginary axis.**



Argand Diagram

Fig. 1.2

e.g. (1) $(1, 2) \equiv 1 + 2i$ (2) $-4+3i \equiv (-4, 3)$

(3) $(0, 0) \equiv 0 + 0i$ (4) $5+0i \equiv (5, 0)$

(5) $(0, -1) \equiv 0 - i$ (6) $-2-2i \equiv (-2, -2)$

A diagram which represents complex numbers by points in a plane with reference to the real and imaginary axes is called **Argand's diagram on complex plane.**

1.5.1 Modulus of z :

If $z = a+ib$ is a complex number then the modulus of z, denoted by $|z|$ or r, is defined as $|z| = \sqrt{a^2 + b^2}$. (From fig. 1.3), point $P(a, b)$ represents the complex number $z = a+ib$.

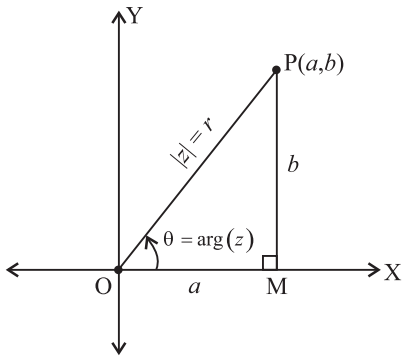
$\therefore r = |z| = \sqrt{a^2 + b^2} = OP$

Hence, modulus of z is the distance of point P from the origin where the point P represents the complex number z in the plane. e.g. For, $z = 4 + 3i$,

Modulus of $z = |z| = \sqrt{16+9} = \sqrt{25} = 5$

1.5.2 Argument of z :

OP makes an angle θ with positive direction of X-axis. θ is called the **argument** or **amplitude** of the complex number $z = a+ib$, denoted by arg(z).



Modulus and Argument of z

Fig. 1.3

$\therefore \sin\theta = \frac{b}{r}$, $\cos\theta = \frac{a}{r}$, $r \neq 0$

$\therefore b = r\sin\theta$, $a = r\cos\theta$

and $\tan\theta = \frac{b}{a}$, if $a \neq 0$

$\therefore \theta = \tan^{-1}\left(\frac{b}{a}\right) = \arg(z)$,

e.g. If $z = 2+2i$ then

$\arg(z) = \theta = \tan^{-1}\left(\frac{2}{2}\right)$

$\therefore \tan^{-1}(1) = \frac{\pi}{4}$

Note : If $\tan x = y$ then its inverse funtion is given by $x = \tan^{-1} y$ or $x = \text{arc}\tan y$

eg:

1) As $\tan\frac{\pi}{6} = \frac{1}{\sqrt{3}}$ then $\tan^{-1} = \left(\frac{1}{\sqrt{3}}\right) = \frac{\pi}{6}$

2) As $\tan\left(-\frac{\pi}{4}\right) = -\tan\frac{\pi}{4} = -1$ then

$\tan^{-1}(-1) = -\frac{\pi}{4}$

1.5.3 Argument of z in different quadrants/axes :

$z = a + ib$	Example	Quadrant/Axis	$\theta = \arg z = \tan^{-1}\left(\frac{b}{a}\right)$, $(0 \le \theta < 2\pi)$	from Example
$a > 0, b = 0$	$z = 3$	On positive real (X) axis	$\theta = 0$	$\theta = 0$
$a > 0, b > 0$	$z = 1 + i$	In quadrant I	$\theta = \tan^{-1}\left(\frac{b}{a}\right)$, $(0 < \theta < \frac{\pi}{2})$	$\theta = \tan^{-1}\left(\frac{1}{1}\right) = \frac{\pi}{4}$
$a = 0, b < 0$	$z = 5i$	On Positive imaginary (Y) axis	$\theta = \frac{\pi}{2}$	$\theta = \frac{\pi}{2}$
$a < 0, b > 0$	$z = -\sqrt{3} + i$	In quadrant II	$\theta = \tan^{-1}\left(\frac{b}{a}\right) + \pi$ $(\frac{\pi}{2} < \theta < \pi)$	$\theta = \tan^{-1}\left(\frac{1}{-\sqrt{3}}\right) + \pi$ $= \frac{-\pi}{6} + \pi = \frac{5\pi}{6})$
$a < 0, b = 0$	$z = -6$	On negative real (X) axis	$\theta = \pi$	$\theta = \pi$

$a < 0,\ b < 0$	$z = -1 - \sqrt{3}i$	In quadrant III	$\theta = \tan^{-1}\left(\frac{b}{a}\right) + \pi$ $(\pi < \theta < \frac{3\pi}{2})$	$\theta = \tan^{-1}\left(\frac{-\sqrt{3}}{-1}\right) + \pi$ $= \frac{\pi}{3} + \pi = \frac{4\pi}{3})$
$a = 0,\ b < 0$	$z = -2i$	On negative imaginary (Y) axis	$\theta = \frac{3\pi}{2}$	$\theta = \frac{3\pi}{2}$
$a > 0,\ b < 0$	$z = 1 - i$	In quadrant IV	$\theta = \tan^{-1}\left(\frac{b}{a}\right) + 2\pi$ $(\frac{3\pi}{2} < \theta < 2\pi)$	$\theta = \tan^{-1}\left(\frac{-1}{1}\right) + 2\pi$ $= \frac{-\pi}{4} + 2\pi = \frac{3\pi}{4})$

SOLVED EXAMPLES

Ex. 1 : If $z = 1+3i$, find the modulus and amplitude of z.

Solution : $z = 1+3i$ here a=1, b=3 and a, b > 0

$\therefore\ |z| = \sqrt{1^2+3^2} = \sqrt{10}$

amp $z = \tan^{-1}\left(\frac{b}{a}\right) = \tan^{-1}\left(\frac{3}{1}\right) = \tan^{-1}(3)$

Ex. 2 : Find the modulus, argument of the complex number $-7 + 24i$.

Solution : let $z = -7+24i$ a = −7, b = 24

$\therefore\ |z| = \sqrt{(7)^2+(24)^2} = \sqrt{625} = 25$

Here,

$\arg z = \tan^{-1}\left(\frac{b}{a}\right) = \tan^{-1}\left(\frac{24}{-7}\right)$

As a < 0, b > 0, θ lies in 2nd quadrant.

Properties of modulus of complex numbers :

If z_1, z_2, z_3 are complex numbers, then

i) $|z| = 0 \Leftrightarrow z = 0$ i.e. Re (z) = Im (z) = 0

ii) $|z| = |-z| = |\overline{z}| = |-\overline{z}|$

iii) $-|z| \le \text{Re}(z) \le |z|;\ -|z| \le \text{Im}(z) \le |z|$

iv) $z\,\overline{z} = |z|^2$

v) $|z_1 z_2| = |z_1|\ |z_2|$

vi) $\left|\frac{z_1}{z_2}\right| = \frac{|z_1|}{|z_2|},\ z_2 \neq 0$

vii) $|z_1 + z_2| \le |z_1| + |z_2|$

viii) $|z_1+z_2|^2 = |z_1|^2 + |z_2|^2 + 2\text{Re}(z_1\overline{z_2})$

ix) $|z_1-z_2|^2 = |z_1|^2 + |z_2|^2 - 2\text{Re}(z_1\overline{z_2})$

x) $|z_1+z_2|^2 + |z_1-z_2|^2 = 2(|z_1|^2 + |z_2|^2)$

xi) $|az_1 - bz_2|^2 + |bz_1 + az_2|^2$

$= (a^2+b^2)\ (|z_1|^2 + |z_2|^2)$ where $a, b \in \mathbb{R}$

Properties of arguments :

1) $\arg(z_1.z_2) = \arg z_1 + \arg z_2$

2) $\arg\left(\frac{z_1}{z_2}\right) = \arg z_1 - \arg z_2$

3) $\arg z + \arg \overline{z} = 0,\ \arg(z\overline{z}) = 0$

4) $\arg \overline{z} = -\arg z = \arg \frac{1}{z}$

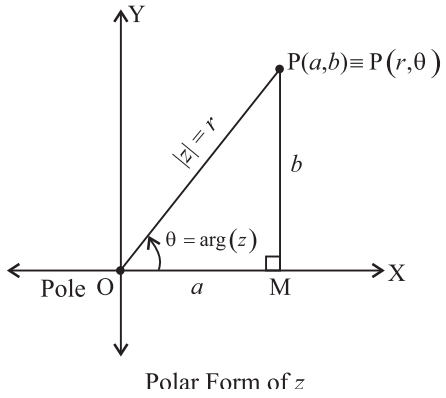
5) If $\arg z = 0$ then z is real

1.5.4 Polar form of a complex number :

Let the complex number $z = a+ib$ be represented by the point P(a, b) (see fig 1.4)

Let $m\angle XOP = \theta = \tan^{-1}\left(\frac{b}{a}\right)$ and $l(OP) = r = \sqrt{a^2+b^2} > 0$, then P$(r,\ \theta)$ are called the Polar Co-ordinates of P.

We call the origin as pole. (figure 1.4)



Polar Form of z

Fig.1.4

As a = $r\cos\theta$, b = $r\sin\theta$

$\therefore$ z = a + ib becomes

z = $r\cos\theta$ + $ir\sin\theta$

$\therefore$ z = $r(\cos\theta + i\sin\theta)$

This is called polar form of complex number z = a + ib

1.5.5 Exponential form :

It is known and can be proved using special series that $e^{i\theta} = \cos\theta + i\sin\theta$

$\therefore$ $z = a + ib = r(\cos\theta + i\sin\theta) = r\,e^{i\theta}$

where $r = |z|$ and $\theta = \arg z$ is called an exponential form of complex number.

Solved Example:

Ex. 1 : Represent the complex numbers

z = 1+i, $\overline{z}$ = 1−i, $-\overline{z}$ = −1+i, −z = −1−i in Argand's diagram and hence find their arguments from the figure.

Solution :

arg z is the angle made by the segment OA with the positive direction of the X-axis. (Fig.1.5)

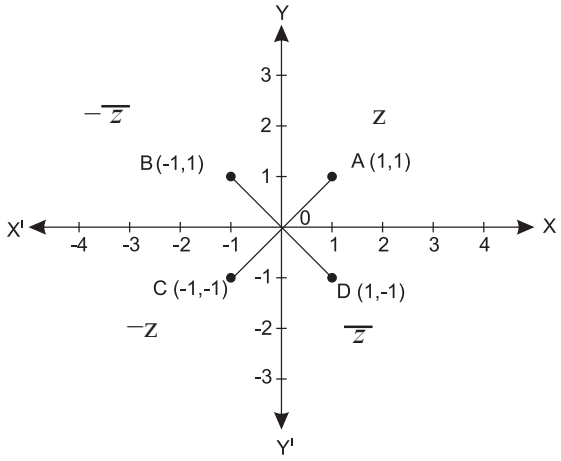


Fig. 1.5

Thus, θ together with r gives the position of the point A in the Argand's diagram.

Hence, from the figure 1.5,

$$\arg z = 45° = \frac{\pi}{4}, \quad \arg(-\overline{z}) = 135° = \frac{3\pi}{4}$$

$$\arg(-z) = 225° = \frac{5\pi}{4}, \quad \arg\overline{z} = 315° = \frac{7\pi}{4}$$

Ex. 2 : Represent the following complex numbers in the polar form and in the exponential form

i) $4+4\sqrt{3}\,i$ ii) -2 iii) $3i$ iv) $-\sqrt{3}+i$

Solution :

i) Let, $z = 4 + 4\sqrt{3}\,i$

a = 4, b = $4\sqrt{3}$

$$r = \sqrt{4^2 + (4\sqrt{3})^2} = \sqrt{16+48} = \sqrt{64} = 8$$

As θ lies in quadrant I

$$\theta = \tan^{-1}\left(\frac{b}{a}\right) = \tan^{-1}\left(\frac{4\sqrt{3}}{4}\right)$$

$$= \tan^{-1}(\sqrt{3}) = \frac{\pi}{3} \text{ or } 60°$$

$\therefore$ The polar form of $z = 4 + 4\sqrt{3}\,i$ is

$z = r(\cos\theta + i\sin\theta)$

$z = 8(\cos 60° + i \sin 60°)$

$= 8\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$

$\therefore$ The exponential form of $z = 8\, e^{i\left(\frac{\pi}{3}\right)}$

ii) Let $z = -2$

$\therefore\ a = -2, b = 0$

Hence, $r = \sqrt{(-2)^2 + 0^2} = \sqrt{4} = 2$

As point $(-2, 0)$ lies on negative real axis

$\theta = \pi^c$ or $180°$

$\therefore$ The polar form of $z = 2(\cos 180° + i \sin 180°)$

$= 2(\cos\pi + i\sin\pi)$

$\therefore$ The exponential form of $z = 2e^{i\pi}$

iii) Let $z = 3i$

$a=0, b=3$

Hence, $r = \sqrt{0^2 + 3^2} = 3$

As point $(0, 3)$ lies on positive imaginary axis

$\theta = \frac{\pi^c}{2}$ or $90°$

$\therefore$ The polar form of $z = 3(\cos 90° + i \sin 90°)$

$= 3\left(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}\right)$

$\therefore$ The exponential form of $z = 3e^{i\frac{\pi}{2}}$

iv) Let, $z = -\sqrt{3} + i$

$\therefore\ a = -\sqrt{3}, \quad b = 1$

$r = \sqrt{(-\sqrt{3})^2 + 1^2} = \sqrt{3+1} = \sqrt{4} = 2$

As $(-\sqrt{3}, 1)$ lies in quadrant II

$\theta = \tan^{-1}\left(\frac{b}{a}\right) = \pi$

$= \tan^{-1}\left(\frac{1}{-\sqrt{3}}\right) + \pi = -\frac{\pi}{6} + \pi$

$= \frac{5\pi}{6}$

$\therefore$ The polar form of

$z = r(\cos\theta + i\sin\theta)$

$z = 2\left(\cos\frac{5\pi}{6} + i\sin\frac{5\pi}{6}\right)$

$\therefore$ The exponential form of

$z = re^{i\theta} = 2\, e^{i\left(\frac{5\pi}{6}\right)}$

Ex. 3 : Express $z = \sqrt{2}.e^{\frac{3\pi}{4}i}$ in the $a + ib$ form.

Solution: $z = \sqrt{2}.e^{\frac{3\pi}{4}i} = r\, e^{i\theta}$

$\therefore\ r = \sqrt{2}, \theta = \frac{3\pi}{4}$

As the polar form of z is

$z = r(\cos\theta + i\sin\theta)$

$= \sqrt{2}\left(\cos\frac{3\pi}{4} + i\sin\frac{3\pi}{4}\right)$

By using allied angles results in trigonometry, we get

$\cos\frac{3\pi}{4} = \cos\left(\pi - \frac{\pi}{4}\right) = -\cos\frac{\pi}{4} = -\frac{1}{\sqrt{2}}$

$\sin\frac{3\pi}{4} = \sin\left(\pi - \frac{\pi}{4}\right) = \sin\frac{\pi}{4} = \frac{1}{\sqrt{2}}$

$\therefore\ z = \sqrt{2}.\left[\frac{-1}{\sqrt{2}} + i\left(\frac{1}{\sqrt{2}}\right)\right]$

$= -1 + i$

Ex. 4 : Express (i) $3.e^{\frac{5\pi}{12}i} \times 4.e^{\frac{\pi}{12}i}$

ii) $\frac{\sqrt{2}\left(\cos\frac{\pi}{12} + i\sin\frac{\pi}{12}\right)}{2\left(\cos\frac{5\pi}{6} + i\sin\frac{5\pi}{6}\right)}$ in $a + ib$ form

Solution: (i) $3.e^{\frac{5\pi}{12}i} \times 4.e^{\frac{\pi}{12}i}$

$= (3 \times 4)\ e^{\left(\frac{5\pi}{12}+\frac{\pi}{12}\right)i}$

$= 12\ e^{\frac{6\pi}{12}i} = 12\ e^{\frac{\pi}{2}i}$

$= 12\left(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}\right)$

$= 12\ (0 + i) = 12i$

(ii) $$\frac{\sqrt{2}\left(\cos\frac{\pi}{12} + i\sin\frac{\pi}{12}\right)}{2\left(\cos\frac{5\pi}{6} + i\sin\frac{5\pi}{6}\right)} = \frac{\sqrt{2}.e^{\frac{\pi}{12}i}}{2e^{\frac{5\pi}{6}i}}$$

$$= \left(\frac{\sqrt{2}}{2}\right)e^{\left(\frac{\pi}{12}-\frac{5\pi}{6}\right)i} = \left(\frac{\sqrt{2}}{2}\right)e^{\left(-\frac{3\pi}{4}\right)i}$$

$$= \left(\frac{\sqrt{2}}{2}\right)\left[\cos\left(-\frac{3\pi}{4}\right) + i\sin\left(-\frac{3\pi}{4}\right)\right]$$

$$= \left(\frac{\sqrt{2}}{2}\right)\left[-\frac{1}{\sqrt{2}} + i\left(-\frac{1}{\sqrt{2}}\right)\right]$$

$$= -\frac{1}{2} - \frac{i}{2}$$

EXERCISE 1.3

1) Find the modulus and amplitude for each of the following complex numbers.

i) $7 - 5i$ ii) $\sqrt{3} + \sqrt{2}i$ iii) $-8 + 15i$

iv) $-3(1-i)$ v) $-4-4i$ vi) $\sqrt{3} - i$

vii) 3 viii) $1 + i$ iv) $1 + i\sqrt{3}$

x) $(1+2i)^2\ (1-i)$

2) Find real values of θ for which $\left(\frac{4+3i\sin\theta}{1-2i\sin\theta}\right)$ is purely real.

3) If $z = 3 + 5i$ then represent the $z, \bar{z}, -z, -\bar{z}$ in Argand's diagram.

4) Express the following complex numbers in polar form and exponential form.

i) $-1+\sqrt{3}\,i$ ii) $-i$ iii) -1 iv) $\frac{1}{1+i}$

v) $\frac{1+2i}{1-3i}$ vi) $\frac{1+7i}{(2-i)^2}$

5) Express the following numbers in the form $x+iy$

i) $\sqrt{3}\,(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6})$

ii) $\sqrt{2}\,(\cos\frac{7\pi}{4} + i\sin\frac{7\pi}{4})$

iii) $7\left(\cos\left(-\frac{5\pi}{6}\right) + i\sin\left(-\frac{5\pi}{6}\right)\right)$

iv) $e^{\frac{\pi}{3}i}$ v) $e^{\frac{-4\pi}{3}i}$ vi) $e^{\frac{5\pi}{6}i}$

6) Find the modulus and argument of the complex number $\frac{1+2i}{1-3i}$.

7) Convert the complex number

$z = \dfrac{i-1}{\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}}$ in the polar form.

8) For $z = 2+3i$ verify the following :

i) $(\bar{z}) = z$ ii) $z\bar{z} = |z|^2$

iii) $(z+\bar{z})$ is real iv) $z - \bar{z} = 6i$

9) $z_1 = 1 + i,\ z_2 = 2 - 3i$. Verify the following :

i) $\overline{z_1 + z_2} = \bar{z_1} + \bar{z_2}$

ii) $\overline{z_1 - z_2} = \bar{z_1} - \bar{z_2}$

iii) $\overline{z_1 . z_2} = \bar{z_1} . \bar{z_2}$

iv) $\overline{\left(\frac{z_1}{z_2}\right)} = \frac{\bar{z_1}}{\bar{z_2}}$